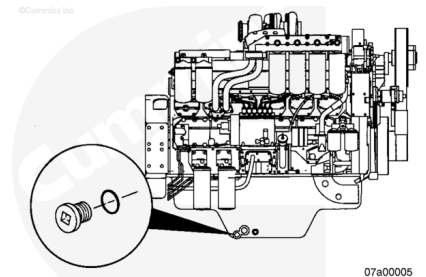


Trainings ▾

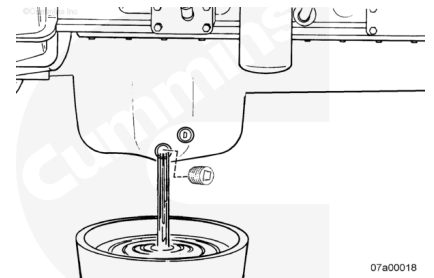
Drain

Change the lubricating oil and filter(s) at the specified oil change interval. Refer to Procedure 018-017 (</qs3/pubsys2/xml/en/procedures/57/57-018-017.html>) Lubricating Oil System in Section 7 for the correct type of oil to use in the engine.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

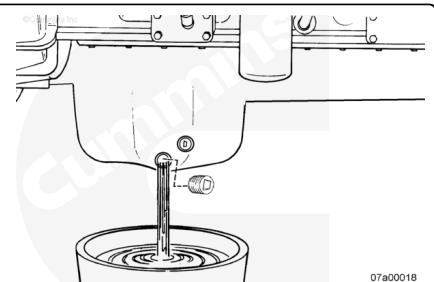
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Operate the engine until the water temperature reaches 60°C [140°F]. Shut off the engine.

Remove the oil drain plug from the bottom of the lubricating oil pan. Do **not** remove the upper plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.

Fittings used in the bottom drain opening of the oil pan other than Cummins specified parts **must not** exceed the following size and weight limits:

- Length: 63.50 mm [2.500 in]
- Diameter: 41.28 mm [1.625 in]
- Weight: 0.363 kg [0.80 lb]



On standby generator applications, Cummins Inc. recommends oil sampling and analysis at the time of oil change to monitor oil contaminant levels.

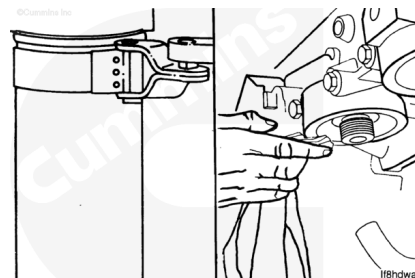
Remove

Use an oil filter wrench, Part Number 3397929, or equivalent, to remove oil filter(s).

Clean the area around the lubricating oil filter head. Clean the gasket surface of the filter head.

The o-ring can stick on the filter head. Make sure the o-ring is removed.

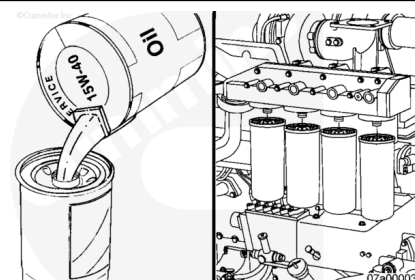
Discard the filters in accordance with local environmental requirements.



Install

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

A shortage of lubrication oil caused by the delay while the filter is pumped full of oil at startup can damage the engine.

Use clean 15W-40 oil to coat the gasket surface of the filter.

Fill the filter with clean 15W-40.

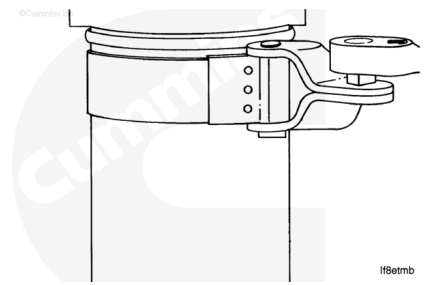
⚠ CAUTION ⚠

Mechanical overtightening of the filter can distort the threads or damage the filter element seal.

Install the filter on the filter head. Tighten the filter +3/4 to 1 turn after the gasket contacts the filter head surface.

Use oil filter wrench, Part Number 3397929, to tighten the filter.

See manufacturer's specifications on tightening the oil filter.

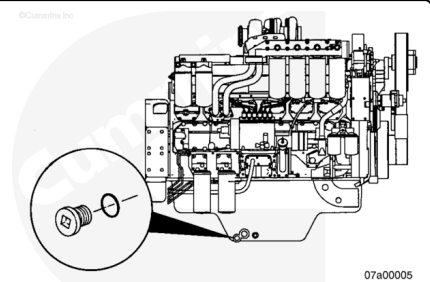


Fill

Clean and check the oil drain plug threads and the seal surface.

Install the oil drain plug in the lubrication oil pan.

Torque Value: 47 n•m [35 ft-lb]



Choose the correct oil for your operating climate.

See Section V for Engine Oil Recommendations and Specifications.



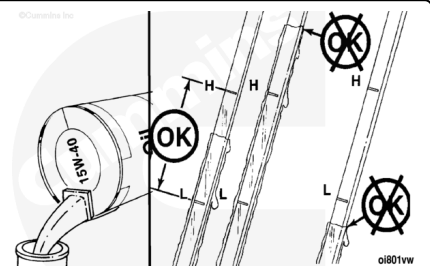
Fill the engine with clean oil to the correct oil level.

Two oil pan capacity options are available.

- Standard (high): 76 liters [20 gal]
- Standard (low): 61 liters [16 gal]
- Option (high): 132 liters [35 gal]
- Option (low): 117 liters [31 gal]

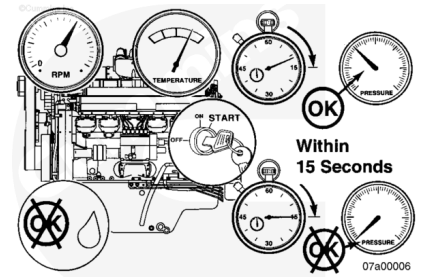
The capacity of each oil filter is listed.

- Full-flow filter: 2.6 liters [0.7 gal]
- Bypass filter: 2.3 liters [0.6 gal]
- Venturi Combo™ filter: 2.5 liters [0.7 gal]



⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to reduce the possibility of engine damage.



Operate the engine at idle speed to inspect for leaks at the filter(s) and the oil drain plug.

Confirm the correct oil level in the oil pan.

Shut off the engine. Wait approximately five minutes for the oil to drain back from the upper parts of the engine to the oil pan.

Check the oil level again. Add oil as necessary to bring the level up to the 'H' (high) mark on the dipstick.

